BINF6970: Assignment 2
Due Friday March 13 by 11:59 PM

Problem 1: (50%)

Background
T cells are continuously generated in the bone marrow to maintain a diverse repertoire capable of responding to pathogens. Following thymic selection, naïve (NAI) T cells migrate to peripheral tissues, where they need to encounter specific antigens and additional signals for clonal expansion and differentiation into effector (EFFE) and memory (MEM) cells. The exact genetic mechanisms governing this differentiation process are not fully understood, and there is ongoing debate as to whether NAI T cells differentiate into EFFEs first, which subsequently become MEM cells (linear model), or if they directly differentiate into both EFFE and MEM cells simultaneously (parallel model).

Consider the data provided in `geneexpression2.rda` where rows are samples from different subjects and different T cell types and the columns are a subset of gene expression measurements on the 156 most differentially expressed genes as studied in (Holmes et al., 2005).

1- Perform PCA on this dataset treating differentially expressed genes as the variables.
2- Visualize the PCs and investigate how they relate to cell and status types.
3- Write down your interpretation considering the conclusions drawn in the article (Holmes et al., 2005)

Note
The row names (e.g. in HEA26_EFFE_1) provide important information:
Hea26_ represents status of the subject and EFFE_1 represents a cell type.
You should read the article to discern cell and status types and overall objective of the analysis.

Problem 2: (50%)

Background
Cytokines storm, a hyperinflammatory state caused by a deregulated immune system, is a key aspect of the novel coronavirus disease, with affected patients showing high levels of pro-inflammatory cytokines, which correlate with disease severity (Vanderkamp et al., 2025). Additional details regarding the role of cytokines in mediating the immune response can be found Table 1 (see Appendix). For this problem, you are provided with a dataset that includes measurements of 28 distinct analytes (cytokines, chemokines, and acute phase proteins) that were obtained from plasma samples of 76 human patients who tested positive for COVID-19.

1- Develop a regularized logistic-regression model via Elastic-Net framework to predict COVID-19 disease progression by analyzing cytokine profiles and patient characteristics.

Task: Your analyses should address the following questions:

1. Which cytokines are most effective at predicting the COVID-19 disease progression in patients?
2. Which cytokines show a significant association with COVID-19 disease severity?
3. How does patient age correlate with COVID-19 disease severity?

Your objective is to develop a regression model to predict COVID-19 disease progression and identify influential covariates. Your analysis should include exploration of the dataset, training the model, and assessing predictive accuracy on a test set using appropriate measures such as accuracy and via AUC-ROC curve analysis involving optimization for sensitivity and specificity.

2- For model evaluation, randomly split the full dataset into a 75% training set and a 25% test set (use seed: 1717 for reproducibility).

3- For model training, use `seq (0, 1, .01)` as the search grid for alpha, i.e. Ridge and Lasso are included in the set of candidate elastic-net models. You may choose to use either "auc" or "deviance" as the accuracy measure (loss function) in `cv.glmnet` at the training stage (only use one of these two measure).

4- Perform two separate sets of analysis by considering 10-fold and 20-fold cross-validation.

5- Discuss the implications of cross-validation schemes (10/20 folds) on your analyses by providing a clear and concise comparison of model selection and predictive performance.

Page limit (Problems 1 and 2 combined): 10 pages

Instructions for Group Work

Assignment 2 is based on teamwork activity involving four members per group. You are required to inform me via email if you intend to change your group formation by the end of **Thursday, March. 5th**.

As well, note the following points:

Group Work: Only ONE member of your group should submit the assignment materials (pdf file and the R code file) to CourseLink.

Files should include the group number and members last names and should be named in the following style:
Group#_last.name1_last.name2. last.name3_ last.name4.pdf
Group#_last.name1_last.name2. last.name3_ last.name4.R

Assessment

1- The evaluation will be based on the following criteria: amount and depth of the analysis, correct use of the statistical methods, correct and logical interpretations of the outcomes of the analyses, clarity and professional appearance of the text, graphics, and R code. The length of the text will not matter, and may be considered even as a negative factor, if overly excessive without a good reason.

2- You are required to include a **Tasks List** and a **Group Work Contribution Table** at the **end** of your main pdf file (solution report). Here are some guidelines to create this table:

   a. The format of the table is given below. Task examples include writing R code for visualization; writing a section of the solution report; interpreting a piece of analysis, developing part of a predictive model, etc.
   b. You should work as group to clearly identify the tasks you performed to arrive at your final solution and list them as Task 1, Task 2, …, Task N. Try to identify major/important tasks only – N=2 (too small) or N=20 (too large) are both unrealistic numbers.
   c. In the table below: CD and (%C) are defined as "contribution details" and "percent contribution", respectively. Your % Cs should sum to 100% for a given task.

3- Your group will be penalized 20% mark if it failed to include the following information.

Tasks List

Task 1: (provide description of the task) Task

2: (provide description of the task)

…

Task N: (provide description of the task)

Group Work Contribution Table

	Student 1	Student 2	Student 3	Student 4
Task 1	CD % C	CD % C	CD % C	CD % C
Task 2	CD % C	CD % C	CD % C	CD % C
…	…	…	…	…
Task N	CD % C	CD % C	CD % C	CD % C

4- Your grade will be penalized if you, as an individual student, failed to collaborate within your group. For example, this can happen when **i)** you simply divide the work in mutually exclusive parts; work individually on your part *only*; and then append your *individual* solution to make it the final *group* solution, and/or **ii)** you do not contribute fairly to the overall assignment and fail to communicate within your group in an effective and professional manner.

Academic Integrity:

Note 1. While you can discuss approaches to problems with other students, the material turned in must be your own solutions. Assignment work is intended to be produced solely by *your group*. Example of academic misconduct include copying solutions from other students/individuals; sharing your solution with other students in your class; soliciting solutions from any other sources such as online websites like chegg.com, etc.

Note 2. Please be reminded that the use of AI tools is **not permitted** for this assignment.

All submissions must reflect your own understanding and original work. Assignments will be reviewed carefully, and any inappropriate use of AI-generated content will be addressed in accordance with academic integrity policies.

If AI assistance is identified, consequences will be applied proportionally based on the extent of the assistance used. If you are unsure about what constitutes acceptable support, please contact me before submitting your work.

Reference:

Holmes, S., He, M., Xu, T., & Lee, P. P. (2005). Memory T cells have gene expression patterns intermediate between naïve and effector. *Proceedings of the National Academy of Sciences of the United States of America*, *102*(15). https://doi.org/10.1073/pnas.0501437102

Vanderkamp, S. G., Niazy, M., Stegelmeier, A. A., Stinson, K. J., Ricker, N., & Bridle, B. W. (2025). Cytokine, chemokine, and acute-phase protein profiles in plasma as correlative biomarkers of clinical outcomes for patients with COVID-19. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-99248-6

Appendix

Table 1: Description of cytokines, chemokines, and acute phase proteins that were measured from plasma samples of COVID-19-positive human patients.

Analytes	Full name & description	Type	Role
IL-6	Interleukin 6	cytokine	Activation of autoimmunity and chronic inflammation.
TNF-α	Tumour Necrosis Factor alpha	cytokine	Inducing necrosis or apoptosis, inhibiting tumorigenesis and viral replication
BTLA	B and T lymphocyte attenuator	cytokine	A co-signaling molecule that induce immunosuppression by inhibiting B and T cell activation and proliferation
LAG3	Lymphocyte activation gene 3	cytokine	Inhibition of cell proliferation, cytokine secretion, and homeostasis
CD27	Cluster of differentiation 27, tumor necrosis factor receptor superfamily	cytokine	Activation of B-cells and prevent tumor growth
TLR-2	Toll-like receptors 2		Improves acquired immunity by inducing antigen presenting cell activation, proinflammatory cytokine
CD28	Cluster of differentiation 28, receptor of the Ig superfamily.	cytokine	Co-stimulatory signals required for T cell activation and survival.
PTX3	Pentraxin 3	Acute phase protein	Regulates complement activation, phagocytosis and tumor promoting inflammation
HVEM	The herpes virus entry mediator (CD270)	cytokine	Regulates the mucosal immune response
GITRL	Glucocorticoid-induced tumor necrosis factor receptor ligand	cytokine	Triggers GITR to induce immune cell signaling, activation, and survival
PD-1	Programmed Cell Death Protein 1	cytokine	Inhibiting immune responses and promoting self-tolerance through modulating the activity of T-cells

PD-L1	Programmed Cell Death Protein Ligand1	cytokine	Preventing T cells, from attacking healthy cells
PD-L2	Programmed Cell Death Protein 1 Ligand 2	cytokine	Negative regulator of T cell activity, inhibiting proliferation, IL-2 production, and IFN-gamma production
CTLA-4	Cytotoxic T-lymphocyte–associated antigen 4	cytokine	Negative regulators of T-cell immune function
CD80 (B7-1)	Cluster of differentiation 80, type I membrane protein	cytokine	Amplification of allergen-specific immune responses through T-T cell interactions
sTNF-RII	Soluble tumor necrosis factor receptor II	cytokine	Activates various inflammatory pathways and induces expression of cyclooxygenase-2
ICOS	Inducible co-stimulator	cytokine	Suppressive antigen-specific T cells
MCP-1/CCL2	Monocyte chemoattractant protein1(CCL2)	chemokine	Regulate migration and infiltration of monocytes/macrophages
CCL4/MIP-1 beta	Beta-chemokines macrophage inflammatory protein	chemokine	Activation of the peripheral blood lymphocytes
CXCL10/IP10/CRG-2	Interferon-inducible protein 10	chemokine	Pro-inflammatory and anti-angiogenic properties
GM-CSF	Granulocyte macrophage colony-stimulating factor	cytokine	Promotes myeloid cell maturation, and dendritic cell differentiation
IL-8/CXCL8	Interleukin 8	chemokine	Recruitment of neutrophils and other immune cells to the site of infection
IL-10	Interleukin 10	cytokine	Limiting host immune response to pathogens by preventing damage and maintaining normal tissue homeostasis
TIM	Transmembrane immunoglobulin and mucin domain	Acute phase protein	Mediates phagocytosis of apoptotic cells by binding to PS exposed on the surface of these cells